



## 2 Real Analysis 2023

### Tutorial 8

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## The ratio and the root tests

**Problem 29** For each of the following series, determine whether it converges or diverges.

- |                                                                    |                                                      |                                                                |
|--------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| (a) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ ,                         | (b) $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$ ,       | (c) $\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$ ,                 |
| (d) $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$ ,                   | (e) $\sum_{n=1}^{\infty} \frac{5^n (n!)^2}{(2n)!}$ , | (f) $\sum_{n=1}^{\infty} \left( \frac{n}{n+1} \right)^{n^2}$ , |
| (g) $\sum_{n=1}^{\infty} 4^n \left( \frac{n}{n+1} \right)^{n^2}$ , | (h) $\sum_{n=1}^{\infty} \frac{n^4}{2^n}$ ,          | (i) $\sum_{n=1}^{\infty} \frac{(2n)!}{n! n^n}$ .               |

## Absolute and conditional convergence

**Problem 30** For a real number  $x$ , we define  $x^+ = \max\{0, x\}$  and  $x^- = \max\{0, -x\}$ .

(a) Show that

$$x = x^+ - x^-, \quad |x| = x^+ + x^-, \quad x^+ = \frac{1}{2}(|x| + x), \quad x^- = \frac{1}{2}(|x| - x).$$

- (b) Let  $\sum_{n=1}^{\infty} a_n$  be a real series. Show that  $\sum_{n=1}^{\infty} a_n$  is absolutely convergent if and only if both series  $\sum_{n=1}^{\infty} a_n^+$  and  $\sum_{n=1}^{\infty} a_n^-$  are convergent.
- (c) Show that if one of the series  $\sum_{n=1}^{\infty} a_n^+$  and  $\sum_{n=1}^{\infty} a_n^-$  converges and the other diverges, then  $\sum_{n=1}^{\infty} a_n$  diverges.
- (d) Show that if  $\sum_{n=1}^{\infty} a_n$  is conditionally convergent, then both series  $\sum_{n=1}^{\infty} a_n^+$  and  $\sum_{n=1}^{\infty} a_n^-$  diverge.

**Problem 31** Prove or disprove.

- (a) The series  $\sum_{n=0}^{\infty} \left(-\frac{2}{3}\right)^n$  can be rearranged so that the sum of the rearranged series is 10.
- (b) The series  $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$  can be rearranged so that the sum of the rearranged series is 10.

**Problem 32** For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent.

$$(a) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2 + 4},$$

$$(b) \sum_{n=0}^{\infty} (-1)^n e^{-n},$$

$$(c) \sum_{n=1}^{\infty} \frac{\sin \sqrt{n}}{\sqrt{n^3 + 2}},$$

$$(d) \sum_{n=1}^{\infty} (-1)^n \frac{\arctan n}{n^2},$$

$$(e) \sum_{n=1}^{\infty} (-1)^n \frac{n^4}{e^n},$$

$$(f) \sum_{n=1}^{\infty} \left( \frac{-n}{2n+1} \right)^n.$$

## Mixed problems on series

**Problem 33** For each of the following series, determine whether it converges or diverges.

$$(a) \sum_{n=1}^{\infty} \frac{\sqrt{n+1} - \sqrt{n}}{n},$$

$$(b) \sum_{n=1}^{\infty} \frac{1}{n} \left( \frac{2}{3} \right)^n,$$

$$(c) \sum_{n=1}^{\infty} n \left( \frac{2}{3} \right)^n,$$

$$(d) \sum_{n=1}^{\infty} \left( \frac{1}{n^2} - \frac{1}{(n+1)^2} \right),$$

$$(e) \sum_{n=1}^{\infty} \frac{1}{2 + 3 \ln n},$$

$$(f) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2 + 3 \ln n}.$$